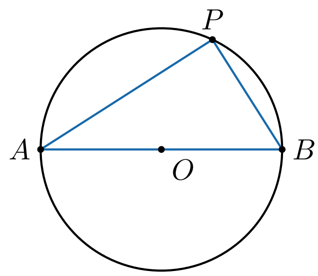
Circle Secrets

The inscribed angle

The big secret. Pick any two points A and B on a circle. The angle you see them at from a rim point P is always exactly *half* the angle you see them at from the center O, and every rim point on the same arc sees them at the *same* angle. Today you measure it, then prove it.

Build 1: the right angle on a diameter

Here is a circle with a diameter AB and one rim point P.



Measure angle APB above with a protractor: __________ degrees.

Draw your own circle below with a diameter AB. Mark four different points P on the rim and measure angle APB at each one. Do they all agree?

Build 2: the rim is half the center

Draw a fresh circle, center O. Mark two points A and B that are *not* a diameter. Draw the center angle AOB. Then pick a point P on the longer arc and draw the rim angle APB.

Center angle $AOB =$ ________, rim angle $APB =$ ________. Is the rim exactly half

the center?

Why is the rim half the center?

Set it up so the line PO runs straight through the center to the far side B (so PB is a diameter). Draw the radius OA to make triangle OAP.

$OA = OP$. Why must these two sides be equal? ______________________________

So the two base angles are equal; call each one t. The center angle AOB is the exterior angle of the triangle at O, so $AOB =$ _________ and the rim angle $APB =$ _________. The rim is half the center.

Push further

1. **Thales for free.** If AB is a diameter, the center angle is the straight line $180°$, so the rim angle must be ________ degrees.
2. **Cyclic quadrilateral (BMC gold).** Put four points on a circle and join them into a quadrilateral. Opposite angles add to ________. Prove it: each corner stands on an arc, and the four arcs cover the whole circle ($360°$).
3. **Competition gem.** Two chords AB and CD cross at a point X inside the circle. Show that triangle AXC and triangle DXB have the same shape (similar). Hint: find two pairs of equal inscribed angles on the same arcs. This gives the length rule $AX \cdot XB = CX \cdot XD$.

Draw the cyclic quadrilateral and the crossing chords here: